

60016 OPERATIONS RESEARCH

Finding a BFS + extensions of linear programming

Lecture 06

Last Lecture

- ▶ Simplex algorithm
 - ▶ Idea
 - ▶ Simplex Tableau
 - ▶ Pivoting
- ▶ Termination
 - ▶ Degenerate BFS
 - ▶ Non-degenerate $\implies$ finite termination

This Lecture

- ▶ Finding an initial BFS
- ▶ Nonlinear problems solvable through LP
 - ▶ Min-max problems
 - ▶ Min-min problems
 - ▶ Fractional linear programming

Finding a BFS - Part I

Simplex starts from a given BFS. How to determine one?

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Two cases:

- solution has $\xi_i \neq 0$ for some $i \implies$ original problem infeasible
- solution has $\xi = 0 \implies$ we have a BFS for original problem

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$$B_A \begin{bmatrix} x_B \\ \xi_B \end{bmatrix} = b, \quad x_B, \xi_B \geq 0 \implies \begin{bmatrix} B_A^1 & B_A^2 \end{bmatrix} \begin{bmatrix} x_B \\ \xi_B \end{bmatrix} = b, \quad x_B, \xi_B \geq 0$$

Therefore we have (almost) found a BFS

Operational details in next exercise session

Important nonlinear problems solvable through LPs

- ▶ Linear programming is useful, but often fairly restrictive.
- ▶ **Nonlinear optimisation problems** arise very commonly in applications.
 - ▶ Euclidean distance, chemical reactions, queueing problems, ...
- ▶ Nonlinearities can affect objective, constraints, or both.
 - ▶ $z = x_1^2$, $z = x \bmod 2$, ...
 - ▶ $|x_1 - x_2| \leq 3$, $x_1 x_2 + x_3 \leq 5$, ...

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- ▶ Nonlinear programs can sometimes be reformulated
 - ▶ as a single LP problem
 - ▶ or as a sequence of LP problems
- ▶ We see the following cases:
 - ▶ min-max models (important for game theory!)
 - ▶ min-min models
 - ▶ fractional linear programming

Min-Max Problems

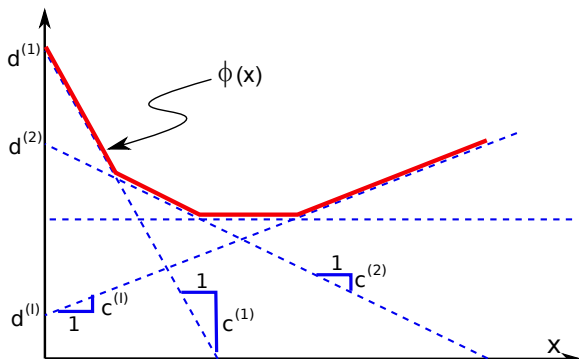
Consider a family of linear functions: $y_i(x) = c(i)^T x + d(i)$.

Set $\phi(x) = \max_{i=1,\dots,l} \{c(i)^T x + d(i)\}$ for $c(i) \in \mathbb{R}^n$, $d(i) \in \mathbb{R}$.

Then,

$$\begin{array}{ll} \min & \phi(x) \\ \text{s.t.} & Ax = b, x \geq 0 \end{array} \quad (\text{MM})$$

is called a **min-max problem**.



Min-Max Problems (cont)

Perhaps surprisingly, Min-Max problems can be solved by LPs.

Theorem. Consider the following linear program:

$$\begin{array}{ll}\min & z \\ \text{s.t.} & z \geq c(i)^T x + d(i) \quad \forall i = 1, \dots, l \\ & Ax = b \\ & x \geq 0, \quad z \text{ free}\end{array} \quad (\text{LP})$$

If (x^*, z^*) is an optimal solution of this LP, then x^* is also an optimal solution of MM, and MM has optimal value $\phi(x^*) = z^*$.

Proof

$$\begin{array}{ll} \min & z \quad (\text{MM}) \\ \text{s.t.} & z = \max_i \{c(i)^T x + d(i)\} \\ & Ax = b \\ & x \geq 0, z \text{ free} \end{array}$$

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1. *The feasible set of (LP) is a superset of feasible set of (MM):*
Indeed, any (x, z) feasible for (MM) must be feasible for (LP)

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$$\min z \quad (\text{MM})$$

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By contradiction, assume this is not the case. Consider $\hat{z} = \max_i \{c(i)^T x^* + d(i)\} \neq z^*$. The point $(\hat{z}, x^*)$ is feasible for (LP) and gives strictly better value $\rightsquigarrow$ contradiction.

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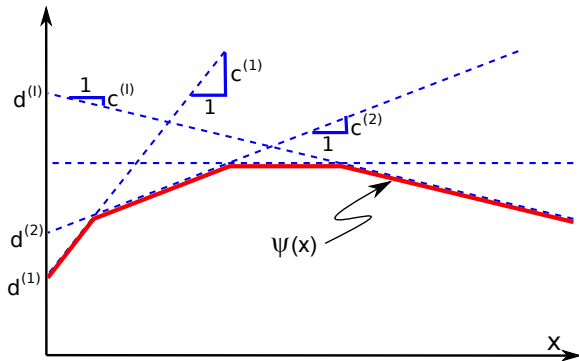
3. *Thus, optimal solution of (LP) must be optimal for (MM).*

Min-Min Problems

Set $\psi(x) = \min_{i=1,\dots,l} \{c(i)^T x + d(i)\}$ for $c(i) \in \mathbb{R}^n$, $d(i) \in \mathbb{R}$. Then,

$$\begin{array}{ll} \min & \psi(x) \\ \text{s.t.} & Ax = b, x \geq 0 \end{array} \quad (\text{MM}') \quad (1)$$

is called a **min-min problem**.



Min-Min Problems (cont)

Theorem: Consider the following linear programs.

$$\begin{array}{ll} \min & z_i = c^T(i)x(i) + d(i) \\ \text{s.t.} & Ax(i) = b \\ & x(i) \geq 0 \end{array} \quad (\text{LP}(i))$$

Let z_i^* be the optimal solution to $\text{LP}(i)$. Let $\text{LP}(j)$ be the LP that has the minimal objective, i.e.,

$$z_j^* = \min_{i=1,\dots,l} z_i$$

and let $x^*(j)$ be its optimal solution. Then $x^*(j)$ is optimal in MM' and $\psi^* = \psi(x^*(j)) = z_j^*$.

Interchangeability of Min-Operations

Lemma: Let X and Y be arbitrary sets, and let $f : X \times Y \rightarrow \mathbb{R}$ be an arbitrary function defined on $X \times Y$. Then,

$$\min_{x \in X} \min_{y \in Y} f(x, y) = \min_{y \in Y} \min_{x \in X} f(x, y) \quad (1)$$

Proof of Min-Min Theorem: The Lemma implies that

$$\left. \begin{array}{ll} \min_{i=1, \dots, I} & \min_x \quad c(i)^T x + d(i) \\ & \text{s.t.} \quad Ax = b \\ & \quad x \geq 0 \end{array} \right\} = \min_{i=1, \dots, I} \text{LP}(i)$$

is equal to

$$\left. \begin{array}{ll} \min_x & \min_{i=1, \dots, I} c(i)^T x + d(i) \\ \text{s.t.} & Ax = b \\ & x \geq 0 \end{array} \right\} = \text{MM}'.$$

Thus, the claim follows.

Fractional Linear Programming

Consider the fractional linear program

$$\min \left\{ \frac{\alpha_0 + \alpha_1 x_1 + \dots + \alpha_n x_n}{\beta_0 + \beta_1 x_1 + \dots + \beta_n x_n} \mid Ax = b; x \geq 0 \right\}. \quad (\text{FLP})$$

We assume that

- ▶ the feasible set of FLP is bounded, i.e.,

$$\exists L > 0 \text{ s.t. } \forall x \text{ satisfying } Ax = b, x \geq 0 \text{ it is } \|x\| \leq L;$$

- ▶ the denominator of the objective function is strictly positive, i.e., $\beta_0 + \beta_1 x_1 + \dots + \beta_n x_n > 0$ for all feasible x .

Homogenisation

- ▶ Introduce **new variables** $y_i \geq 0$, $i = 1, \dots, n$ and $y_0 > 0$.
- ▶ Setting $x_i = \frac{y_i}{y_0}$, $i = 1, \dots, n$, we can **homogenise** the fractional linear program as follows.

$$\left. \begin{array}{ll} \min & \frac{\alpha_0 y_0 + \alpha_1 y_1 + \alpha_2 y_2 + \dots + \alpha_n y_n}{\beta_0 y_0 + \beta_1 y_1 + \beta_2 y_2 + \dots + \beta_n y_n} \\ \text{s.t.} & b_i y_0 - \sum_{j=1}^n a_{ij} y_j = 0 \quad \forall i = 1, \dots, m \\ & y_0 > 0, y_1 \geq 0, \dots, y_n \geq 0 \end{array} \right\} \quad (\text{HFLP})$$

Homogenisation

- ▶ For any $(y_0, y_1, \dots, y_n)$ feasible in HFLP, $\lambda(y_0, y_1, \dots, y_n)$ with $\lambda > 0$ is **also feasible** and has the **same objective value**.
- ▶ For each $(y_0, y_1, \dots, y_n)$ we can then find a λ such that

$$\beta_0 y_0 + \dots + \beta_n y_n = 1.$$

and the scaled point will have identical objective. Thus we can restrict our attention to these points and still find an optimal solution. In other words:

- ⇒ The denominator in the objective of HFLP can always be **normalised** to unity.

Normalised Problem

Solve the **normalised problem**

$$\min \quad \alpha_0 y_0 + \alpha_1 y_1 + \alpha_2 y_2 + \dots + \alpha_n y_n$$

$$\text{s.t.} \quad \beta_0 y_0 + \sum_{j=1}^n \beta_j y_j = 1$$

$$b_i y_0 - \sum_{j=1}^n a_{ij} y_j = 0 \quad \forall i = 1, \dots, m$$

$$y_0 > 0, y_1 \geq 0, \dots, y_n \geq 0$$

where the first constraint forces the **denominator of the objective** of HFLP to be **equal to 1**.

$\Rightarrow$ **This is an LP!**

Constructing a Solution for FLP

- ▶ Denote by $(y_0^*, y_1^*, \dots, y_n^*)$ the **optimal solution** of the **normalised problem**.
- ▶ Then, $(\frac{y_1^*}{y_0^*}, \dots, \frac{y_n^*}{y_0^*})$ is an **optimal solution** for FLP.
- ▶ Note: this construction **only works** if $y_0^* \neq 0$.
- ▶ However, we can show (**next slide**) that under the assumptions y_0 cannot be zero at optimality.

Relaxing the Lower Bound on y_0

- ▶ Let us now assume to use in HFLP $y_0 \geq 0$ instead of $y_0 > 0$.
- ▶ Suppose $y_0^* = 0$ and set $y^* = (y_1^*, \dots, y_n^*)^T$. Since $b_i y_0^* = 0$

$$Ay^* = 0, y^* \geq 0$$

- ▶ Here $y^* \neq 0$ or otherwise the denominator of the objective in both HFLP and FLP would not be strictly positive as assumed.
- ▶ We now note that, if x is feasible in FLP, then $x + \lambda y^*$ is also feasible in FLP $\forall \lambda > 0$ since given that $Ay^* = 0$ it is

$$A(x + \lambda y^*) = Ax + \lambda Ay^* = b,$$

- ▶ This would contradict that FLP has a bounded feasible set, so y_0^* cannot be zero and we can use $y_0 \geq 0$ in HFLP.